EMBEDDED VISION PORTFOLIO ARTIFACT

EuroSAT Benchmark, Fine-Tuning, and Deployment Readiness

A recruiter-friendly engineering page for evaluating modern edge backbones on top-down imagery, adapting the strongest candidates with transfer learning, and exporting deployment artifacts for ONNX and TensorRT-style inference stacks.

Head-reset transfer baseline Recorded on CUDA

GitHub Repository [https://github.com/KenanTrivedi/efficient-vision-benchmark]

Technical Report [https://github.com/KenanTrivedi/efficient-vision-benchmark/blob/main/REPORT.md]

PDF Portfolio [assets/EuroSAT_Edge_Vision_Benchmark_Portfolio.pdf]

WHY THIS MATTERS

ImageNet intuition alone is not enough for aerial robotics. This project makes the pipeline legible: benchmark the hardware trade-offs, fine-tune on the actual visual domain, and prepare the winning model for edge deployment.

GitHub Pages [https://kenantrivedi.github.io/efficient-vision-benchmark/]

README [https://github.com/KenanTrivedi/efficient-vision-benchmark/blob/main/README.md]

Quantum Systems role [https://career.quantum-systems.com/o/ai-software-engineer-mfd]

MODELS BENCHMARKED

9

Architectures in the baseline sweep

BEST BASELINE ACCURACY

FastViT-MCI4

13.41 % on the held-out test split

FASTEST CPU MODEL

MobileNetV4-Small

15.62 ms median CPU latency

TOP FINE-TUNED MODEL

ConvNeXt-Tiny

99.14% top-1 after supervised adaptation

METHODOLOGY

Interpret the baseline correctly

Each pretrained backbone is adapted to EuroSAT by replacing the final classifier with a fresh 10-class layer and evaluating before training.

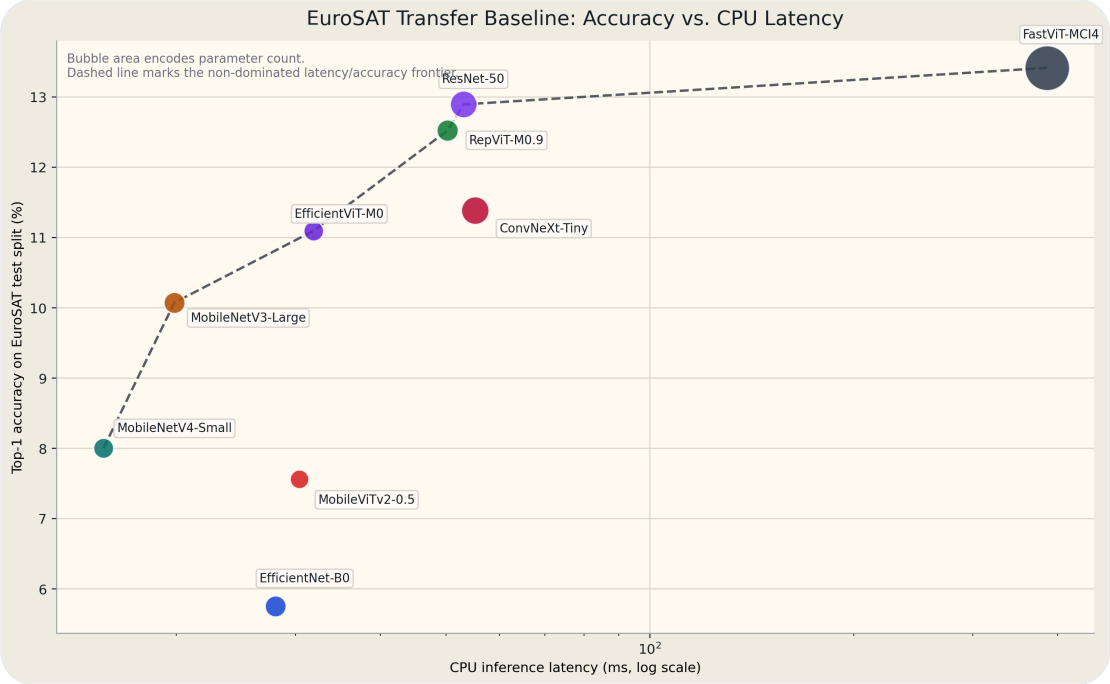
ROLE ALIGNMENT

Why this maps cleanly to Quantum Systems

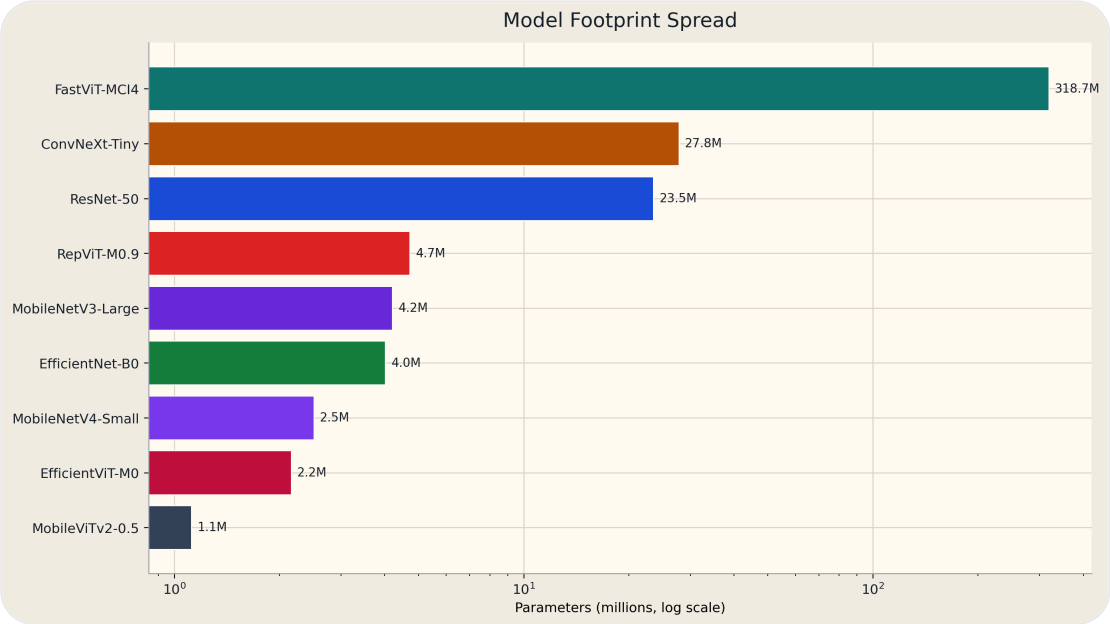
- Preparing and selecting data, training and validating models, and deploying them on embedded UxV platforms.
- Optimization for constrained hardware, including quantization, pruning, and distillation.
- Practical deployment on edge accelerators such as NVIDIA Jetson and TensorRT-backed pipelines.

BASELINE VISUALS

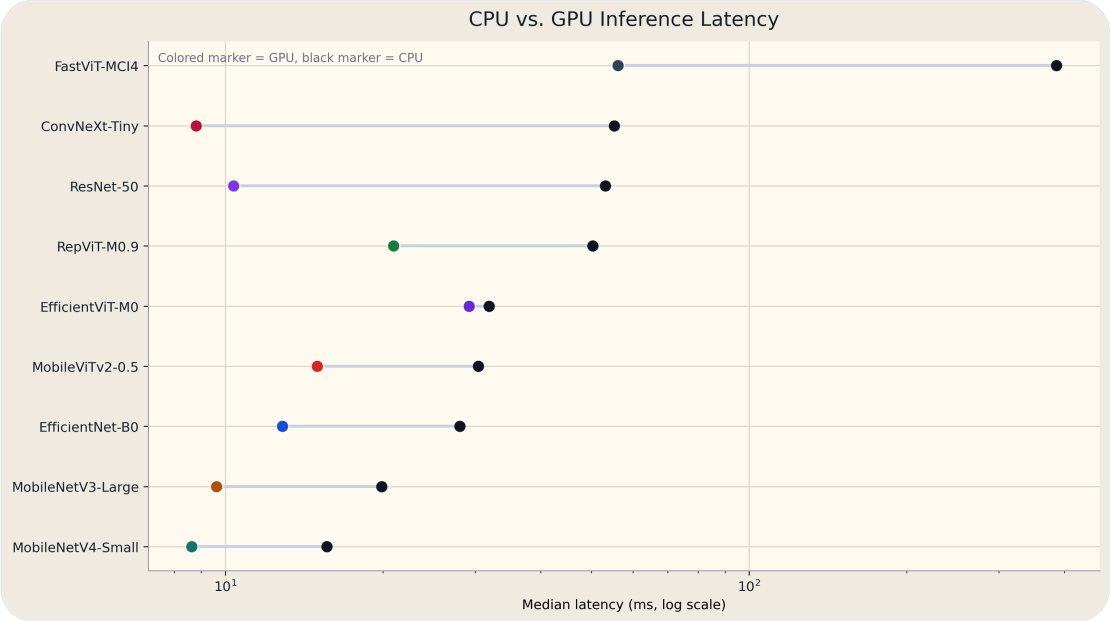
Cleaner figures for README, GitHub Pages, and PDF export



Latency versus accuracy with a Pareto frontier and bubble sizes scaled by parameter count.



Log-scale footprint chart so large outliers no longer flatten the useful part of the portfolio.



CPU versus GPU latency split, which is the first deployment question after model selection.

BENCHMARK LEDGER

Normalized baseline view

SORTED BY CPU LATENCY

Model	Top-1	CPU	GPU	Params	MACs	Batch-1
MobileNetV4-Small	8.00%	15.62 ms	8.62 ms	2.5M	188M	138.7 img/s
MobileNetV3-Large	10.07%	19.87 ms	9.62 ms	4.2M	230M	97.7 img/s
EfficientNet-B0	5.75%	28.02 ms	12.84 ms	4.0M	409M	75.8 img/s

Model	Top-1	CPU	GPU	Params	MACs	Batch-1
MobileViTv2-0.5	7.56%	30.39 ms	14.98 ms	1.1M	472M	81.7 img/s
EfficientViT-M0	11.09%	31.88 ms	29.19 ms	2.2M	79M	41.2 img/s
RepViT-M0.9	12.52%	50.31 ms	20.95 ms	4.7M	839M	57.2 img/s
ResNet-50	12.89%	53.10 ms	10.36 ms	23.5M	4130M	114.6 img/s
ConvNeXt-Tiny	11.38%	55.22 ms	8.80 ms	27.8M	4488M	101.7 img/s
FastViT-MCI4	13.41%	385.84 ms	56.13 ms	318.7M	22112M	16.5 img/s

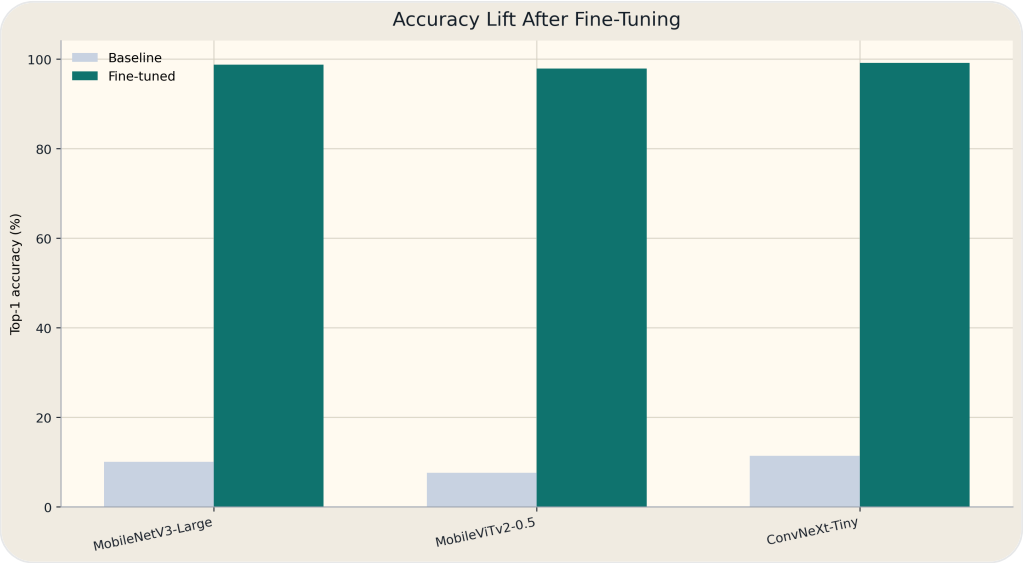
FINE-TUNING OUTCOME

Supervised adaptation results

DEPLOYMENT TARGET: MOBILENETV3-LARGE — FASTEST MODEL WITHIN 1 PERCENTAGE POINT OF THE TOP FINE-TUNED ACCURACY.

3 fine-tuned models completed. The strongest checkpoint is ready for deployment export.

Model	Top-1	Gain
ConvNeXt-Tiny	99.14%	+87.76 pts
MobileNetV3-Large	98.74%	+88.67 pts
MobileViTv2-0.5	97.90%	+90.34 pts



Baseline versus fine-tuned accuracy on the held-out EuroSAT test split.

SELECTED NEXT TARGETS

Recommended fine-tune seeds

1. MobileViTv2-0.5 — strong candidate for supervised adaptation.
2. MobileNetV3-Large — strong candidate for supervised adaptation.
3. ConvNeXt-Tiny — strong candidate for supervised adaptation.

DEPLOYMENT FOLLOW-THROUGH

What the export pipeline adds

The export pipeline packages MobileNetV3-Large into ONNX, prepares calibration data, and attempts TensorRT engine builds when local tooling is available.

DEPLOYMENT ARTIFACTS

Export status and generated files

MOBILENETV3-LARGE • ONNX RUNTIME AVAILABLE • TENSORRT TOOLING MISSING

AVAILABLE

FP32 ONNX

results/deployment/mobilenet_v3_large/mobilenet_v3_large_fp32.onnx • 16.07 MB

AVAILABLE

ONNX validation

max abs diff 0.003182

AVAILABLE

Calibration data

results/deployment/mobilenet_v3_large/calibration_data.npz • 147 MB

OK

INT8 ONNX

results/deployment/mobilenet_v3_large/mobilenet_v3_large_int8_qdq.onnx • 4.25 MB • onnxruntime-static-qdq

SKIPPED

TensorRT FP16 engine

trtexec was not found on PATH.

SKIPPED

TensorRT INT8 engine

trtexec was not found on PATH.

DELIVERABLES

Portfolio links recruiters can use immediately

CODEBASE

GitHub repository

Source, scripts, and tracked results.

[https://github.com/KenanTrivedi/efficient-vision-benchmark]

WRITE-UP

Technical report

Methodology, caveats, and role alignment.

[https://github.com/KenanTrivedi/efficient-vision-benchmark/blob/main/REPORT.md]

PROJECT PAGE

GitHub Pages site

Clean, printable summary with figures and links.

[https://kenantrivedi.github.io/efficient-vision-benchmark/]

PDF

Recruiter-ready export

Printable packet with highlighted hyperlinks.

[assets/EuroSAT_Edge_Vision_Benchmark_Portfolio.pdf]

EuroSAT • train 18900, val 4050, test 4050 • generated 14/03/2026, 23:04:21

Repository **[https://github.com/KenanTrivedi/efficient-vision-benchmark]**

Report **[https://github.com/KenanTrivedi/efficient-vision-benchmark/blob/main/REPORT.md]**

Role posting **[https://career.quantum-systems.com/o/ai-software-engineer-mfd]**